



三角函数知识点结构图

- 角的定义及弧度: $180^\circ = \pi \text{ rad}; 1^\circ = \frac{\pi}{180} \text{ rad}$

(Definition and radian measure of angles: $180^\circ = \pi \text{ rad}; 1^\circ = \frac{\pi}{180} \text{ rad}$)

- 三角函数的定义及关系:

(Definition and relations of trigonometric functions:)

$$\circ \sin \alpha = \frac{y}{r}; \cos \alpha = \frac{x}{r}; \tan \alpha = \frac{y}{x}; \cot \alpha = \frac{x}{y}$$

$$(\sin \alpha = \frac{y}{r}; \cos \alpha = \frac{x}{r}; \tan \alpha = \frac{y}{x}; \cot \alpha = \frac{x}{y})$$

$$\circ \sec \alpha = \frac{r}{x}; \csc \alpha = \frac{r}{y}$$

$$(\sec \alpha = \frac{r}{x}; \csc \alpha = \frac{r}{y})$$

$$\circ \text{基本关系式: } \sin^2 \alpha + \cos^2 \alpha = 1; \tan \alpha = \frac{\sin \alpha}{\cos \alpha}; \tan \alpha \cdot \cot \alpha = 1$$

$$(\text{Basic relations: } \sin^2 \alpha + \cos^2 \alpha = 1; \tan \alpha = \frac{\sin \alpha}{\cos \alpha}; \tan \alpha \cdot \cot \alpha = 1)$$

- 三角函数的诱导公式: $2k\pi + \alpha$ 、 $\pi - \alpha$ 、 $\pi + \alpha$ 、 $-\alpha$ 的三角函数值(含拓展公式 $\frac{\pi}{2} \pm \alpha$)

(Induction formulas of trigonometric functions: Trigonometric values of $2k\pi + \alpha$, $\pi - \alpha$, $\pi + \alpha$, $-\alpha$ (including extended formulas $\frac{\pi}{2} \pm \alpha$))

- 正余弦、正余切函数的图像和性质

(Graphs and properties of sine, cosine, tangent, and cotangent functions)

4.1 角的定义及弧度

(4.1 Definition and Radian Measure of Angles)

一、基础知识(I. Basic Knowledge)

(一)任意角(1. Arbitrary Angle)

定义:平面内一条射线绕着它的端点从一个位置旋转到另一个位置所形成的图形。(Definition: A figure formed by a ray in a plane rotating from one position to another around its endpoint.)

1. 分类:

(Classification:)

- 正角:逆时针方向旋转形成的角(例: 110°)

(Positive angle: Angle formed by counterclockwise rotation (example: 110°))

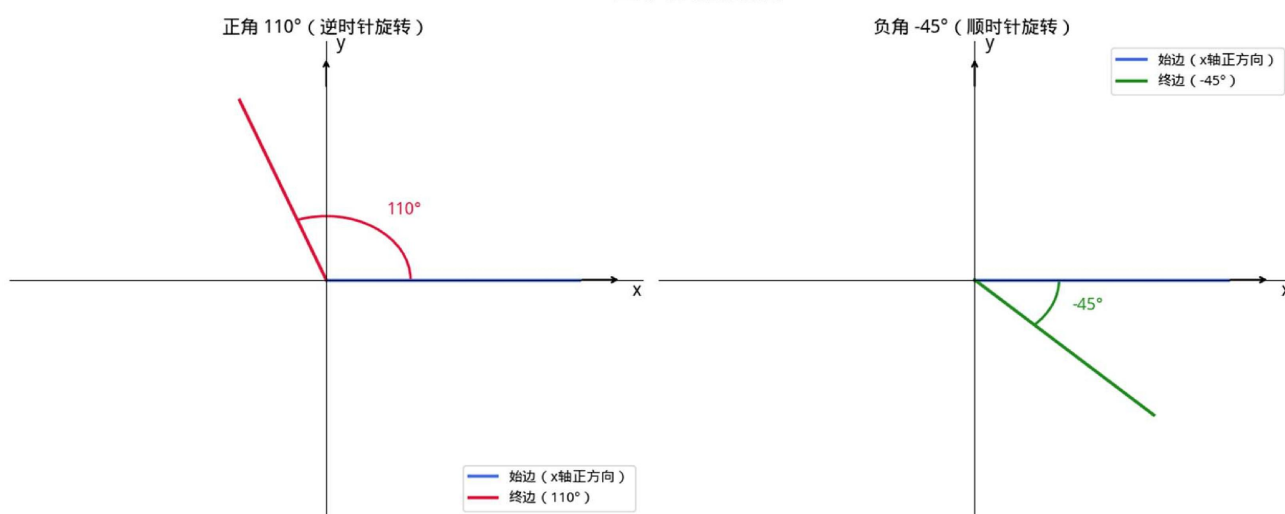
- 负角:顺时针方向旋转形成的角(例: -45°)

(Negative angle: Angle formed by clockwise rotation (example: -45°))

- 零角:没有任何旋转的角

(Zero angle: Angle with no rotation at all)

正角与负角示意图





(二)角度和弧度(2. Degrees and Radians)

1. 换算关系:

(Conversion relations:)

- $180^\circ = \pi \text{ rad}$

$$(180^\circ = \pi \text{ rad})$$

- $1^\circ = \frac{\pi}{180} \text{ rad}$

$$(1^\circ = \frac{\pi}{180} \text{ rad})$$

(三)象限角(3. Quadrant Angles)

- 定义:角的顶点和原点重合,始边和 x 轴正方向重合,角的终边落在第几象限,就称该角为第几象限角。

(Definition: When the vertex of an angle coincides with the origin, and its initial side coincides with the positive direction of the x -axis, the angle is called a "quadrant angle" corresponding to the quadrant where its terminal side lies.)

2. 各象限角的范围($k \in \mathbb{Z}$):

(Range of each quadrant angle

($k \in \mathbb{Z}$):)

- 第一象限角: $k \cdot 360^\circ < \alpha <$

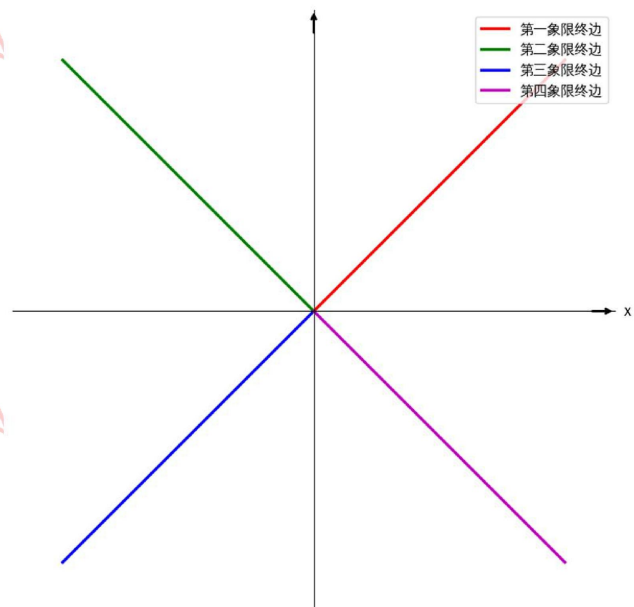
$$k \cdot 360^\circ + 90^\circ$$

(1st quadrant angle: $k \cdot 360^\circ <$

$$\alpha < k \cdot 360^\circ + 90^\circ)$$

- 第二象限角: $k \cdot 360^\circ + 90^\circ <$

象限角分类示意图



$$\alpha < k \cdot 360^\circ + 180^\circ$$

(2nd quadrant angle: $k \cdot 360^\circ + 90^\circ \leq \alpha < k \cdot 360^\circ + 180^\circ$)

- 第三象限角: $k \cdot 360^\circ + 180^\circ \leq \alpha < k \cdot 360^\circ + 270^\circ$

(3rd quadrant angle: $k \cdot 360^\circ + 180^\circ < \alpha < k \cdot 360^\circ + 270^\circ$)

- 第四象限角: $k \cdot 360^\circ + 270^\circ < \alpha < k \cdot 360^\circ + 360^\circ$

(4th quadrant angle: $k \cdot 360^\circ + 270^\circ < \alpha < k \cdot 360^\circ + 360^\circ$)

(四)锐角和钝角(4. Acute and Obtuse Angles)

1. 锐角:大于 0° 且小于 90° 的角。

(Acute angle: Angle greater than 0° and less than 90° .)

2. 钝角:大于 90° 且小于 180° 的角。

(Obtuse angle: Angle greater than 90° and less than 180° .)



4.2 三角函数的定义及关系

(4.2 Definition and Relations of Trigonometric Functions)

一、基础知识(I. Basic Knowledge)

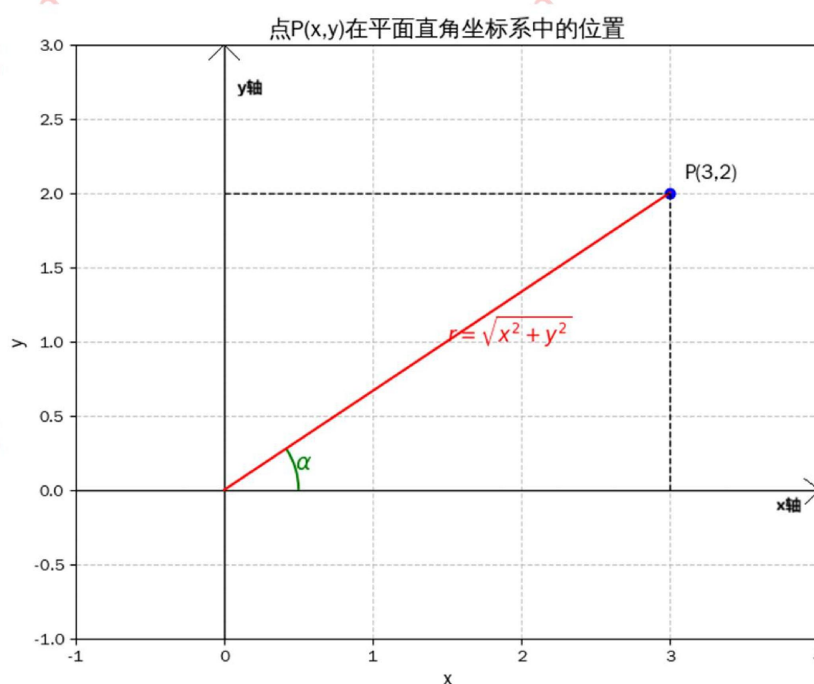
(一)三角函数的定义(1. Definition of Trigonometric Functions)

设平面内一点 $P(x, y)$, 点 P 到原点的距离为 $r = \sqrt{x^2 + y^2} (r > 0)$, 则角 α 的六个三角函数定义为:

Given a point $P(x, y)$ in a plane, with the distance from P to the origin being

$r = \sqrt{x^2 + y^2} (r > 0)$, the six trigonometric functions of angle α are defined as:

1. 正弦: $\sin \alpha = \frac{y}{r}$ (Sine: $\sin \alpha = \frac{y}{r}$)
2. 余弦: $\cos \alpha = \frac{x}{r}$ (Cosine: $\cos \alpha = \frac{x}{r}$)
3. 正切: $\tan \alpha = \frac{y}{x} (x \neq 0)$ (Tangent: $\tan \alpha = \frac{y}{x} (x \neq 0)$)
4. 余切: $\cot \alpha = \frac{x}{y} (y \neq 0)$ (Cotangent: $\cot \alpha = \frac{x}{y} (y \neq 0)$)
5. 正割: $\sec \alpha = \frac{r}{x} (x \neq 0)$ (Secant: $\sec \alpha = \frac{r}{x} (x \neq 0)$)
6. 余割: $\csc \alpha = \frac{r}{y} (y \neq 0)$ (Cosecant: $\csc \alpha = \frac{r}{y} (y \neq 0)$)

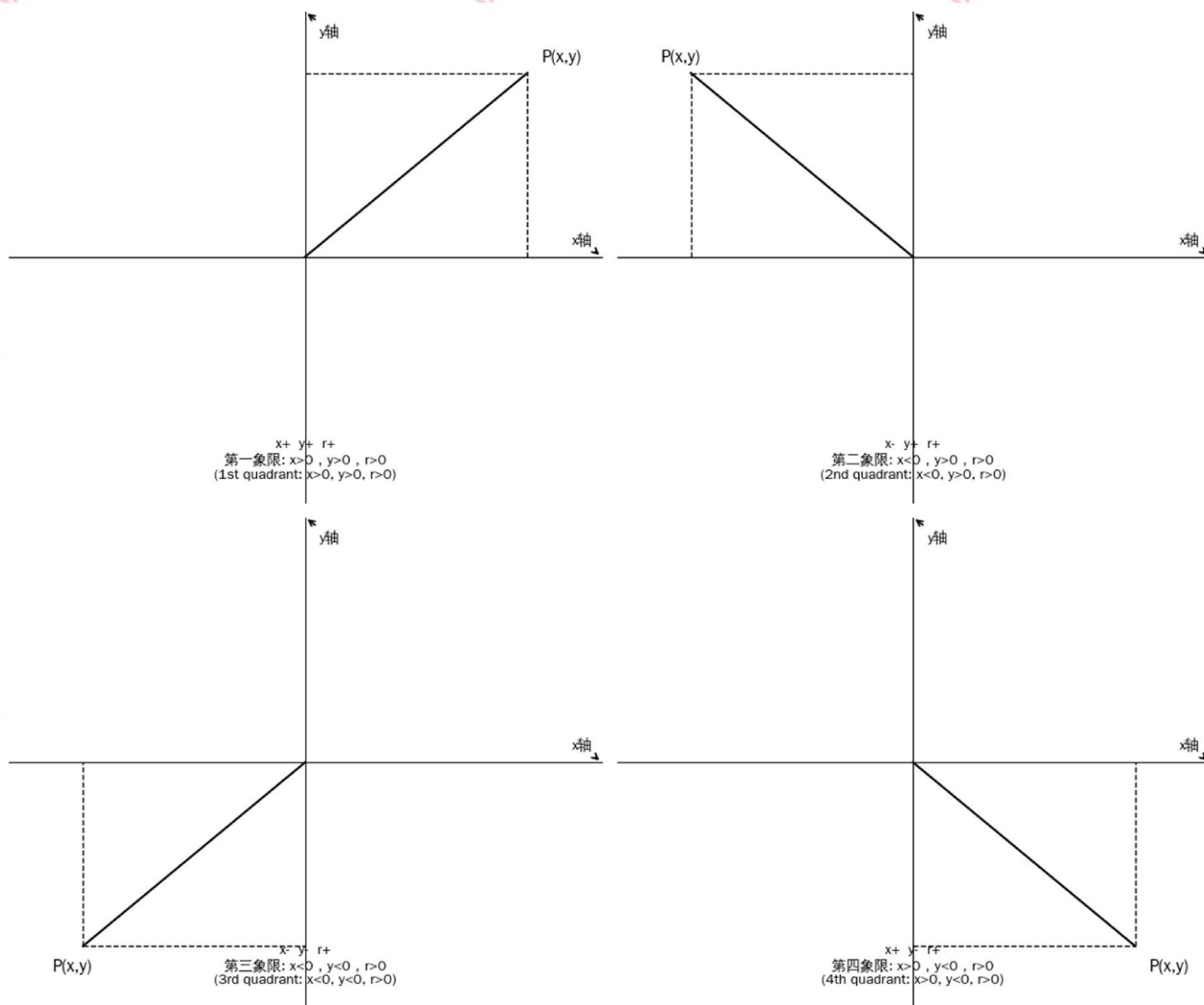


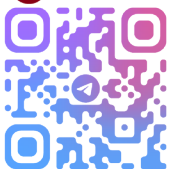
(二)三角函数在各象限的正负符号(2. Positive/Negative Signs of Trigonometric Functions in Each Quadrant)

1. 坐标与 r 的正负规律:

(Positive/negative rules of coordinates and r :)

- 第一象限: $x > 0, y > 0, r > 0$ (1st quadrant: $x > 0, y > 0, r > 0$)
- 第二象限: $x < 0, y > 0, r > 0$ (2nd quadrant: $x < 0, y > 0, r > 0$)
- 第三象限: $x < 0, y < 0, r > 0$ (3rd quadrant: $x < 0, y < 0, r > 0$)
- 第四象限: $x > 0, y < 0, r > 0$ (4th quadrant: $x > 0, y < 0, r > 0$)





2. 三角函数正负口诀:一全正, 二正弦, 三正切, 四余弦

(Mnemonic for positive/negative trigonometric functions: All positive in Q1, only sine positive in Q2, only tangent positive in Q3, only cosine positive in Q4)

- 第一象限: $\sin\alpha$ 、 $\cos\alpha$ 、 $\tan\alpha$ 、 $\cot\alpha$ 、 $\sec\alpha$ 、 $\csc\alpha$ 均为正

(Q1: $\sin\alpha$, $\cos\alpha$, $\tan\alpha$, $\cot\alpha$, $\sec\alpha$, $\csc\alpha$ are all positive)

- 第二象限: 仅 $\sin\alpha$ 、 $\csc\alpha$ 为正

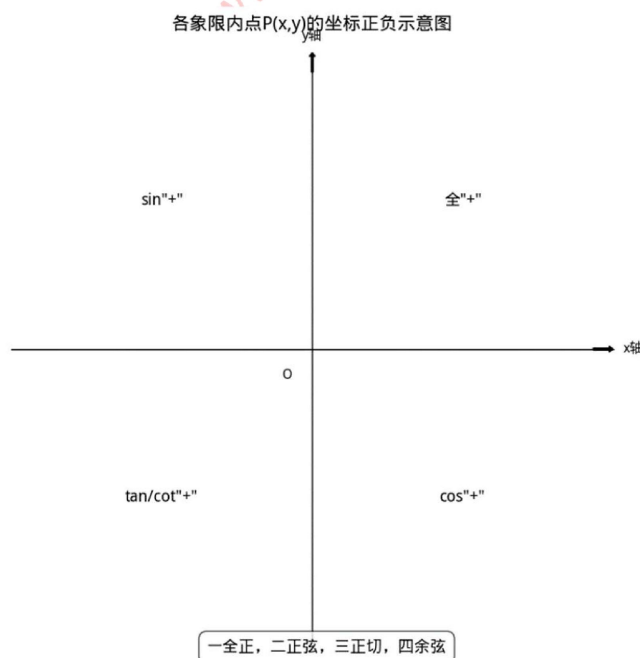
(Q2: Only $\sin\alpha$ and $\csc\alpha$ are positive)

- 第三象限: 仅 $\tan\alpha$ 、 $\cot\alpha$ 为正

(Q3: Only $\tan\alpha$ and $\cot\alpha$ are positive)

- 第四象限: 仅 $\cos\alpha$ 、 $\sec\alpha$ 为正

(Q4: Only $\cos\alpha$ and $\sec\alpha$ are positive)



(三)特殊角的三角函数值(3. Trigonometric Values of Special Angles)

角度	$\sin\alpha$	$\cos\alpha$	$\tan\alpha$	$\cot\alpha$
0°	0	1	0	无(None)
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$
90°	1	0	无(None)	0
180°	0	-1	0	无(None)

(四)同角三角函数的基本关系式(4. Basic Relations of Trigonometric Functions with the Same Angle)

1. $\sin^2\alpha + \cos^2\alpha = 1$

$(\sin^2\alpha + \cos^2\alpha = 1)$

2. $\tan\alpha = \frac{\sin\alpha}{\cos\alpha} (\cos\alpha \neq 0)$

$(\tan\alpha = \frac{\sin\alpha}{\cos\alpha} (\cos\alpha \neq 0))$

3. $\tan\alpha \cdot \cot\alpha = 1$ ($\tan\alpha$ 、 $\cot\alpha$ 均有意义)

$(\tan\alpha \cdot \cot\alpha = 1 \text{ (where } \tan\alpha \text{ and } \cot\alpha \text{ are defined)})$



4.3 三角函数的诱导公式

(4.3 Induction Formulas of Trigonometric Functions)

一、基础知识(I. Basic Knowledge)

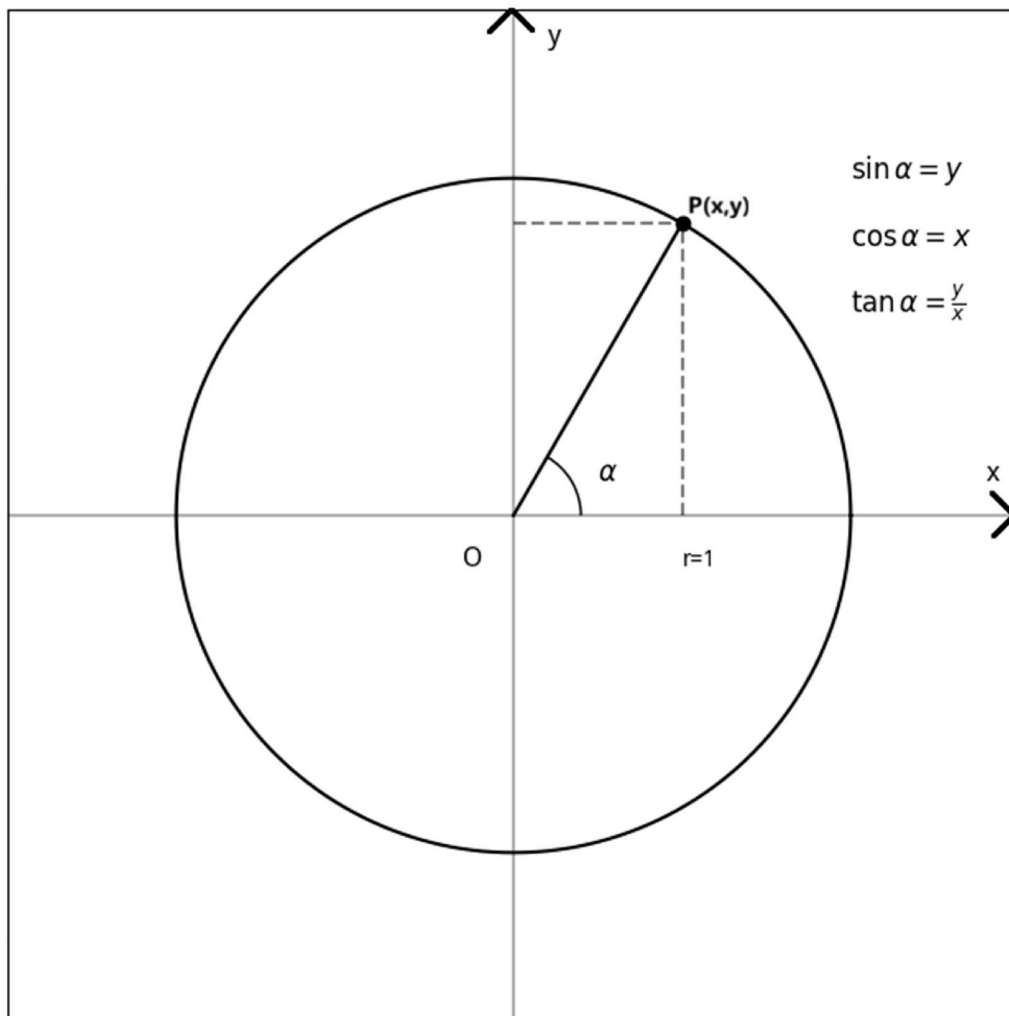
(一)单位圆(1. Unit Circle)

1. 定义:以原点为圆心, 以单位长度为半径的圆。

(Definition: A circle with the origin as the center and unit length as the radius.)

2. 单位圆中三角函数的表示: $\sin \alpha = y$, $\cos \alpha = x$, $\tan \alpha = \frac{y}{x} (x \neq 0)$

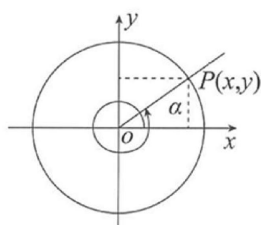
(Expression of trigonometric functions in the unit circle: $\sin \alpha = y$, $\cos \alpha = x$, $\tan \alpha = \frac{y}{x} (x \neq 0)$)



(二)核心诱导公式($k \in \mathbb{Z}$)(2. Core Induction Formulas ($k \in \mathbb{Z}$))

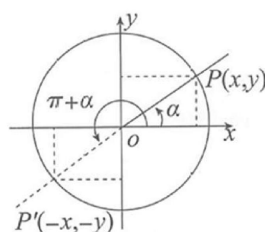
公式类别	三角函数关系(Trigonometric Relations)
公式一($2k\pi + \alpha$)	$\sin(2k\pi + \alpha) = \sin\alpha; \cos(2k\pi + \alpha) = \cos\alpha; \tan(2k\pi + \alpha) = \tan\alpha$
公式二($\pi + \alpha$)	$\sin(\pi + \alpha) = -\sin\alpha; \cos(\pi + \alpha) = -\cos\alpha; \tan(\pi + \alpha) = \tan\alpha$
公式三($-\alpha$)	$\sin(-\alpha) = -\sin\alpha; \cos(-\alpha) = \cos\alpha; \tan(-\alpha) = -\tan\alpha$
公式四($\pi - \alpha$)	$\sin(\pi - \alpha) = \sin\alpha; \cos(\pi - \alpha) = -\cos\alpha; \tan(\pi - \alpha) = -\tan\alpha$

公式一(α 和 $2k\pi + \alpha$ 三角函数的关系)



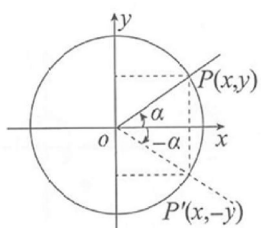
$$\begin{aligned}\sin(2k\pi + \alpha) &= \sin\alpha \quad (k \in \mathbb{Z}) \\ \cos(2k\pi + \alpha) &= \cos\alpha \quad (k \in \mathbb{Z}) \\ \tan(2k\pi + \alpha) &= \tan\alpha \quad (k \in \mathbb{Z})\end{aligned}$$

公式二(α 和 $\pi + \alpha$ 三角函数的关系)



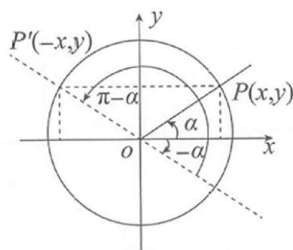
$$\begin{aligned}\sin(\pi + \alpha) &= -\sin\alpha \\ \cos(\pi + \alpha) &= -\cos\alpha \\ \tan(\pi + \alpha) &= \tan\alpha\end{aligned}$$

公式三(α 和 $-\alpha$ 三角函数的关系)



$$\begin{aligned}\sin(-\alpha) &= -\sin\alpha \\ \cos(-\alpha) &= \cos\alpha \\ \tan(-\alpha) &= -\tan\alpha\end{aligned}$$

公式四(α 和 $\pi - \alpha$ 三角函数的关系)



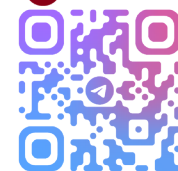
$$\begin{aligned}\sin(\pi - \alpha) &= \sin\alpha \\ \cos(\pi - \alpha) &= -\cos\alpha \\ \tan(\pi - \alpha) &= -\tan\alpha\end{aligned}$$

(三)诱导公式记忆口诀(3. Mnemonic for Induction Formulas)

函数值不变, 正负看象限(将 α 视为锐角, 判断原角所在象限, 确定三角函数值的正负)(Function value remains unchanged; positive/negative depends on the quadrant (treat α as an acute angle, determine the quadrant of the original angle, and confirm the sign of the trigonometric value))

(四)拓展诱导公式(自学内容)(4. Extended Induction Formulas (Self-study Content))

$$\begin{aligned}\sin\left(\frac{\pi}{2} - \alpha\right) &= \cos\alpha; \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha; \\ \sin\left(\frac{\pi}{2} + \alpha\right) &= \cos\alpha; \cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha\end{aligned}$$



4.4 正弦函数和余弦函数的图像和性质

(4.4 Graphs and Properties of Sine and Cosine Functions)

一、基础知识(I. Basic Knowledge)

(一)正弦函数 $y = \sin x$ 与余弦函数 $y = \cos x$ 的核心性质(1. Core Properties of

Sine Function $y = \sin x$ and Cosine Function $y = \cos x$)

函数 (Function)	$y = \sin x$	$y = \cos x$
图像 (Graph)		
定义域 (Domain)	$\mathbb{R}$	$\mathbb{R}$
值域 (Range)	$[-1, 1]$	$[-1, 1]$
周期(T) (Period (T))	$2k\pi (k \in \mathbb{Z}, \text{最小正周期} 2\pi)$ $2k\pi (k \in \mathbb{Z}, \text{minimum positive period } 2\pi)$	$2k\pi (k \in \mathbb{Z}, \text{最小正周期} 2\pi)$ $2k\pi (k \in \mathbb{Z}, \text{minimum positive period } 2\pi)$
奇偶性 (Parity)	奇函数($\sin(-x) = -\sin x$) Odd function ($\sin(-x) = -\sin x$)	偶函数($\cos(-x) = \cos x$) Even function ($\cos(-x) = \cos x$)
单调性 (Monotonicity)	增区间: $(2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}) (k \in \mathbb{Z})$; 减区间: $(2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}) (k \in \mathbb{Z})$	增区间: $(2k\pi - \pi, 2k\pi) (k \in \mathbb{Z})$; 减区间: $(2k\pi, 2k\pi + \pi) (k \in \mathbb{Z})$
	Increasing interval: $(2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}) (k \in \mathbb{Z})$; Decreasing interval: $(2k\pi + \frac{\pi}{2}, 2k\pi + \frac{3\pi}{2}) (k \in \mathbb{Z})$	Increasing interval: $(2k\pi - \pi, 2k\pi) (k \in \mathbb{Z})$; Decreasing interval: $(2k\pi, 2k\pi + \pi) (k \in \mathbb{Z})$

(二)周期函数的定义(2. Definition of Periodic Function)

1. 一般地, 对于函数 $f(x)$, 如果存在一个非零常数 T , 使得当 x 取定义域内的每一个值时, 都有 $f(x+T) = f(x)$, 那么函数 $f(x)$ 叫作周期函数。

(In general, for a function $f(x)$, if there exists a non-zero constant T such that $f(x+T) = f(x)$ for every x in the domain, then $f(x)$ is called a periodic function.)

2. 非零常数 T 叫作这个函数的周期。

(The non-zero constant T is called the period of the function.)

(三)形如 $y = A\sin(\omega x + \varphi)$ 或 $y = A\cos(\omega x + \varphi)$ ($A > 0$, $\omega > 0$)的性质(3.

Properties of Functions in the Form $y = A\sin(\omega x + \varphi)$ or $y = A\cos(\omega x + \varphi)$ ($A > 0$, $\omega > 0$)

1. 最小正周期: $T = \frac{2\pi}{|\omega|}$

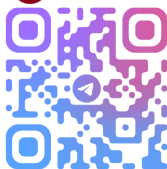
(Minimum positive period: $T = \frac{2\pi}{|\omega|}$)

2. 最值:最大值 $y_{max} = A$, 最小值 $y_{min} = -A$

(Maximum and minimum values: Maximum value $y_{max} = A$, minimum value $y_{min} = -A$)

3. 注意:最小正周期由 x 前的系数 ω 决定, 最值由系数 A 决定。

(Note: The minimum positive period is determined by the coefficient ω in front of x , and the maximum/minimum values are determined by the coefficient A .)



4.5 正切函数和余切函数的图像和性质

(4.5 Graphs and Properties of Tangent and Cotangent Functions)

一、基础知识(I. Basic Knowledge)

(一)正切函数 $y = \tan x$ 与余切函数 $y = \cot x$ 的核心性质(1. Core Properties of

Tangent Function $y = \tan x$ and Cotangent Function $y = \cot x$)

函数 (Function)	$y = \tan x$	$y = \cot x$
图像 (Graph)		
定义域 (Domain)	$\{x \mid x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$	$\{x \mid x \neq k\pi, k \in \mathbb{Z}\}$
值域(Range)	$\mathbb{R}$	$\mathbb{R}$
周期(T)	$k\pi (k \in \mathbb{Z}, \text{ 最小正周期}\pi)$	$k\pi (k \in \mathbb{Z}, \text{ 最小正周期}\pi)$
(Period (T))	$k\pi (k \in \mathbb{Z}, \text{ minimum positive period } \pi)$	$k\pi (k \in \mathbb{Z}, \text{ minimum positive period } \pi)$
奇偶性(Parity)	奇函数($\tan(-x) = -\tan x$)	奇函数($\cot(-x) = -\cot x$)
	Odd function ($\tan(-x) = -\tan x$)	Odd function ($\cot(-x) = -\cot x$)
单调性 (Monotonicity)	在 $(k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}) (k \in \mathbb{Z})$ 内 单调递增	在 $(k\pi, k\pi + \pi) (k \in \mathbb{Z})$ 内单调递 减
	Monotonically increasing in $(k\pi - \frac{\pi}{2}, k\pi + \frac{\pi}{2}) (k \in \mathbb{Z})$	Monotonically decreasing in $(k\pi, k\pi + \pi) (k \in \mathbb{Z})$

(二)形如 $y = A\tan(\omega x + \varphi)$ 或 $y = A\cot(\omega x + \varphi)$ ($A > 0, \omega > 0$)的性质(2.

Properties of Functions in the Form $y = A\tan(\omega x + \varphi)$ or $y =$

$A\cot(\omega x + \varphi)$ ($A > 0, \omega > 0$)

1. 最小正周期: $T = \frac{\pi}{|\omega|}$

(Minimum positive period: $T = \frac{\pi}{|\omega|}$)

2. 定义域:

(Domain:)

• $y = A\tan(\omega x + \varphi): \omega x + \varphi \neq \frac{\pi}{2} + k\pi (k \in \mathbb{Z})$

($y = A\tan(\omega x + \varphi): \omega x + \varphi \neq \frac{\pi}{2} + k\pi (k \in \mathbb{Z})$)

• $y = A\cot(\omega x + \varphi): \omega x + \varphi \neq k\pi (k \in \mathbb{Z})$

($y = A\cot(\omega x + \varphi): \omega x + \varphi \neq k\pi (k \in \mathbb{Z})$)